

PRACTICAL - 6

HAMMING CODE

AIM:-

To write a program to implement error detection and correcting using Hamming code. and to test our data input streams and verify error correction feature.

CODE:-

```
data = input("Enter data bits: ")
n = len(data)
r = 0
while 2**r < n+r+1:
    r += 1
print("Redundancy bits:", r)
total = n+r
h = [0] * (total+1)
j = 0
for i in range(total, 0, -1):
    if (i & (i-1)) != 0:
        h[i] = int(data[j])
        j += 1
    for p in range(r):
        pos = 2**p
        parity = 0
        for i in range(1, total+1):
            if i & pos:
                parity = parity ^ h[i]
        h[pos] = parity
print("Data to be transmitted: ", end=" ")
for i in range(total, 0, -1):
    print(h[i], end=" ")
print()

change = input("Is there any change in transmitted bit? (Yes/No): ")
if change.lower() == "yes":
    received = input("Enter received data: ")
    j = 0
    for i in range(total, 0, -1):
        h[i] = int(received[j])
        j += 1
    error_position = 0
```



```

for p in range(1, total+1):
    if i * pos:
        parity = parity ^ h[i]
    if parity:
        error_position += pos
    print("Error position:", error_position)
    if error_position != 0:
        h[error_position] = h[error_position] ^ 1
    print("Corrected data: ", end=" ")
    for i in range(total, 0, -1):
        print(h[i], end=" ")
    print()
else:
    print("No error detected.")

```

Output:-

① Enter Redundancy data bits: 1101110

Redundancy bits: 4

Data to be transmitted: 110011100

Is there any change in transmitted bit? (yes/no): Yes

Enter received data: 11001101001

Error position: 5

Corrected data: 1100111001

② Enter data bits: 1011011

Redundancy bits: 4

Data to be transmitted: 1010101011

Is there any change in transmitted bit? (yes/no): Yes

Enter received data: 0010101011

Error position: 11

Corrected data: 1010101011

→ If: No

• No error detected.

RESULT:-

Thus, the Hamming code was implemented successfully for error detection and correction.